
BIOLOGY

9700/34

Paper 3 Advanced Practical Skills 2

May/June 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

PUBLISHED**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mark scheme abbreviations

;	separates marking points
/	alternative answers for the same point
R	reject
A	accept (for answers correctly cued by the question, or by extra guidance)
AW	alternative wording (where responses vary more than usual)
<u>underline</u>	actual word given must be used by candidate (grammatical variants accepted)
max	indicates the maximum number of marks that can be given
ora	or reverse argument
mp	marking point (with relevant number)
ecf	error carried forward
I	ignore
AVP	alternative valid point

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Question	Answer	Marks
1(a)(i)	shows 4 concentrations of ethanol as $7.5 + 3.75 + 1.875 + 0.9375 + \%$; shows transfer of 20 cm^3 of ethanol solution from beaker to beaker ; shows addition of 20 cm^3 water to each beaker ;	3
1(a)(ii)	1 heading for percentage concentration of ethanol ; 2 heading for colour + pH ; 3 records colours for all concentrations at both 3 and 9 minutes ; 4 records pH values for all concentrations at both 3 and 9 minutes ; 5 pH values recorded as whole numbers ;	5
1(a)(iii)	correct estimate based on candidates results ;	1
1(a)(iv)	correct statement to reject or accept hypothesis ; correct reference to data from (a)(ii) ;	2
1(a)(v)	difficulty in judgement of colour and matching to pH ;	1
1(b)(i)	label on x-axis time / minutes + label on y-axis volume of CO_2 / arbitrary units (au) ; scale on x-axis is 2 to 2 cm + y-axis is 0.2 to 2 cm + labelled each 2 cm ; correct plotting of five points with a small cross or dot in circle ; line sharp and joined point to point ;	4
1(b)(ii)	collects correct values from graph for volumes at 10 minutes and 7 minutes ; shows changes in volume divided by 3 ;	2

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Question	Answer	Marks
1(b)(iii)	thermostatically controlled water-bath ; at least 5 temperatures ; mass / volume of yeast the same or concentration of glucose the same ;	3

Question	Answer	Marks
2(a)(i) 2(a)(ii) 2(a)(iii)	draws yeast cells of different shapes and sizes ; draws only whole cells ; draws only 3 cells per box ;	3
2(a)(iv)	correct annotation describing an observable difference between Y and M ; correct annotation describing an observable difference between Y and I ;	2
2(b)	minimum cell size + lines thin and continuous ; only 5 whole cells drawn ; at least two yeast cells with inclusions ; cells drawn as in Fig. 2.2 ;	4
2(c)(i)	minimum size + no cells ; draws at least 2 layers of tissue in B1 ; draws tissues in correct proportions ; draws inner layer of B1 crinkled ;	4

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Question	Answer	Marks
2(c)(ii)	B1 walls thicker than B2 ; B1 innermost layer is crinkled, B2 innermost layer smooth ; B2 has a larger lumen than B1 ;	3
2(c)(iii)	B1 + thick(er) wall / more muscle tissue / AW ;	1
2(c)(iv)	shows measurement of scale bar in mm + multiplication by 1000 + division by 535 ; displays answer as a whole number ;	2